

Habitation prototypes

Check out the habitation prototypes made by private spaceflight companies that NASA commissioned. <https://digit.in/may20-73>



Competition killing the Gateway

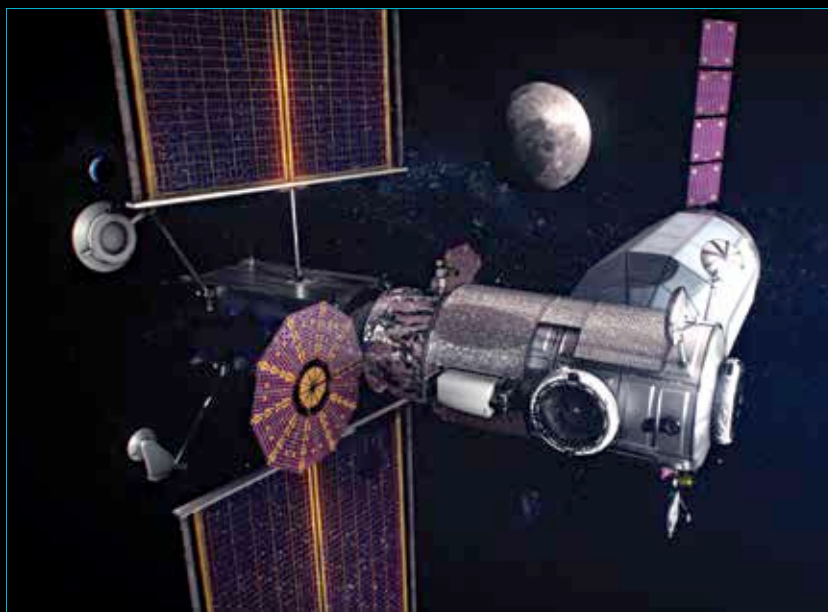
Competition of the Gateway mission may be trying to get the project killed off. <https://digit.in/may20-74>

the US utilisation module, which has some extra habitation capacity. This module is also scheduled for a 2023 launch. Think of it as an expanded closet. This module will also have an external robotic arm attached, to help with the operations of the space station. The arm will be used to inspect spacecraft, help them berth, and also manage the science payloads, or experiments. After that is the habitation module, which is both the office space and the living quarters for the astronauts. Like most humans right now, astronauts

provide additional storage space for the missions when required. A small sample return spacecraft can also dock directly to the habitation module, with the explicit purpose of gathering and returning material from the lunar surface. The two habitation modules are scheduled for deployment in 2024 and 2025, while the robotic arm and the first cargo supply craft are both lined up for 2024 launches. After that, attached to the second habitation module is an airlock for the crew, which can accommodate docking spacecraft.

2018. One of the harshest critics was Apollo 11 astronaut, Buzz Aldrin, who called the idea "absurd". Aldrin found it unnecessary to break up future crewed missions to the Moon into two parts, one to the gateway and another to the surface. Other experts have stated that there are no new technologies that can be validated in orbit around the Moon, that has not already been demonstrated by the ISS around Earth. Additionally, the real science and opportunity for extracting necessary resources are on the surface of the Moon itself, which would, perhaps, serve as a better location for a launchpad for deep space missions into the outer solar system. Such a base would also be more suitable for any experiments that scientists might want to conduct. The Gateway is also criticised for being too expensive, and providing very little benefit in return for the investment. One of the few ways an orbiting platform around the Moon can be useful is to save propellant by not requiring spaceships to actually land on the Moon to refuel. This would mean that such a Gateway would benefit humans if it was a fuel depot, where the propellant manufactured on the surface of the Moon, and processed there, is lifted into orbit for other spacecraft to refuel. However, such a setup is still much more into the future than the planned completion of the Gateway by 2026.

In March this year, NASA made announcements about some of the changes to the Artemis missions, and one of them was reducing the reliance of the program on the Lunar Gateway. NASA still clarified that it is committed to go ahead with building the Gateway, but only that the space station was very likely to fall behind schedule, and that the Artemis missions would proceed as planned anyway. The ongoing work towards the Moon mission has been affected by the coronavirus outbreak, and there may be other long term effects and changes to the plans as well. 



Credit: NASA

Phase 1 of the gateway, as planned for completion by 2024

also have to work from home, and in isolation. The habitation module will be pressurised, and have environment control, life support systems, fire detection, as well as water storage and distribution. It will also contain the facilities for healthcare, sanitation, and exercise for the astronauts. These might seem like a lot of things to pack in a single studio apartment, but bear in mind that in space, all the walls are floors. There are two habitation modules, chained up one after the other. The cargo shuttles to resupply the spacecraft will dock directly with the habitation module, and contain consumables for the crew. The cargo craft can

One of the benefits of the platform is that it will have an almost continuous view of the Earth. The planned mission duration of the Gateway is about 15 years.

THE CRITICISMS

The Lunar gateway concept is, however, not without criticism. Initially, the reception for the concept was enthusiastic by the space community at the International Astronautical Congress in October 2018. The idea was, however, not received well, to put it mildly, by a panel of experts during a meeting by the US National Space Council's Users' Advisory Group, a month later in November